

An orbital primal-dual theorem for finite permutation actions

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Abstract

For any finite group acting on a nonempty finite set, we identify the sharp atom response under marginal-preserving probability perturbations with a rational orbital linear program and its minimum-oscillation dual. Rational optimizers exist, and positive optima are attained by disjoint central laws and sharp small perturbations of the uniform law. Neither faithfulness nor transitivity is assumed. All principal conclusions have complete Lean proofs.

1 The exact finite-action theorem

Let G be a finite group acting (not necessarily transitively or faithfully) on a finite nonempty set Ω , with $e \in G$ its identity. Write u_G for the uniform law on G , and let $\mathcal{V}$ be the real vector space of signed functions $v : G \rightarrow \mathbb{R}$ satisfying

$$\sum_{g \in G} v(g) = 0, \quad \sum_{\substack{g \in G \\ g(x)=y}} v(g) = 0 \quad \text{for all } x, y \in \Omega. \quad (1)$$

Let $\mathcal{O}_1, \dots, \mathcal{O}_r$ be the orbitals of the action, i.e. the orbits of G on $\Omega \times \Omega$ under simultaneous relabeling. Define their *orbital displacement counts*

$$F_j(g) = \#\{x \in \Omega : (x, g(x)) \in \mathcal{O}_j\}. \quad (2)$$

Each F_j is constant on conjugacy classes of G . Let $C_1, \dots, C_s$ be those conjugacy classes, with $C_1 = \{e\}$, and put

$$M_{ji} = \frac{1}{|C_i|} \sum_{g \in C_i} F_j(g), \quad M \in \mathbb{Q}^{r \times s}. \quad (3)$$

For $\mathcal{V} \neq \{0\}$, define

$$\mathcal{C}(G, \Omega) = \sup_{\substack{v \in \mathcal{V} \\ v \neq 0}} \frac{|v(e)|}{\frac{1}{2} \sum_g |v(g)|}. \quad (4)$$

For $\mathcal{V} = \{0\}$, set $\mathcal{C}(G, \Omega) = 0$. The supremum is finite and $0 \leq \mathcal{C}(G, \Omega) \leq 1$.

Theorem (exact orbital primal-dual characterization). The following four quantities are all equal:

$$\begin{aligned}
\mathcal{C}(G, \Omega) &= \max_{\substack{P, Q \text{ probability laws on } G \\ P\{g:gx=y\}=Q\{g:gx=y\} \ \forall x,y}} (P(e) - Q(e)) \\
&= \max_{\substack{p, q \in \mathbb{R}_{\geq 0}^s \\ \mathbf{1}^T p = \mathbf{1}^T q = 1, \ M p = M q}} (p_1 - q_1) \\
&= \min_{\lambda \in \mathbb{R}^r} \left[\max_{1 \leq i \leq s} (\mathbf{1}_{\{i=1\}} - \lambda^T M_{\cdot i}) - \min_{1 \leq i \leq s} (\mathbf{1}_{\{i=1\}} - \lambda^T M_{\cdot i}) \right] \\
&= \min_{\varphi \in \text{span}_{\mathbb{R}}\{F_1, \dots, F_r\}} \text{osc}_{g \in G} (\mathbf{1}_{\{g=e\}} - \varphi(g)).
\end{aligned} \tag{5}$$

Both finite LPs in the middle have **rational optimal solutions**, so $\mathcal{C}(G, \Omega) \in \mathbb{Q}$. If $\mathcal{C}(G, \Omega) > 0$, there exist two **conjugation-invariant, disjointly supported** probability laws P, Q with identical one-point marginals and $P(e) - Q(e) = \mathcal{C}(G, \Omega)$. Hence for all sufficiently small $\delta \geq 0$,

$$\nu_\delta = u_G + \delta(P - Q) \tag{6}$$

is a probability law with exactly the same one-point marginals as u_G and

$$\|\nu_\delta - u_G\|_{\text{TV}} = \delta, \quad \nu_\delta(e) - u_G(e) = \mathcal{C}(G, \Omega)\delta. \tag{7}$$

For an arbitrary probability law ν whose marginals match u_G , and any $\sigma \in G$,

$$\left| \nu(\sigma) - \frac{1}{|G|} \right| \leq \mathcal{C}(G, \Omega) \|\nu - u_G\|_{\text{TV}}, \tag{8}$$

with this constant optimal whenever it is positive. Left translation moves the sharp construction from e to any prescribed σ .

Proof. First formulate a finite **unreduced** LP. Choose P, Q as nonnegative probability vectors indexed by G , impose their equality of all one-point marginals, and maximize $P(e) - Q(e)$. Call its optimum γ .

For any feasible pair, $v = P - Q \in \mathcal{V}$, $\|v\|_{\text{TV}} \leq 1$, so $P(e) - Q(e) \leq \mathcal{C}(G, \Omega)$. Conversely, for any nonzero $v \in \mathcal{V}$, its Jordan parts v_+, v_- each have mass $d = \|v\|_{\text{TV}} > 0$, and the probability measures $P = v_+/d$, $Q = v_-/d$ have identical marginals. Exchanging them if necessary, they give objective $|v(e)|/d$. Hence $\gamma = \mathcal{C}(G, \Omega)$. If $\mathcal{V} = \{0\}$, the two quantities are both zero.

The unreduced primal is a rational finite LP. Its dual has one unrestricted multiplier λ_{xy} for each marginal equality and two normalization multipliers a, b . Write

$$\phi(g) = \sum_{x, y \in \Omega} \lambda_{xy} \mathbf{1}_{\{g(x)=y\}}.$$

Dual feasibility is exactly

$$a \geq \mathbf{1}_{\{g=e\}} - \phi(g), \quad b \geq -\mathbf{1}_{\{g=e\}} + \phi(g) \quad \text{for all } g \in G.$$

Minimizing $a + b$ for fixed ϕ gives $\max_g (\mathbf{1}_{\{g=e\}} - \phi(g)) - \min_g (\mathbf{1}_{\{g=e\}} - \phi(g))$, its oscillation. The primal is feasible (take $P = Q = u_G$) and bounded, so **finite-dimensional LP strong duality** gives equality with the minimum oscillation. The LP has rational coefficients and a finite attained optimum; standard rational Gaussian elimination at a basic feasible optimum gives rational primal and dual certificates.

Next average an optimal primal pair under simultaneous conjugation by $h \in G$: replace P, Q by the averages of their pushforwards under $g \mapsto hgh^{-1}$. Equal one-point marginals remain equal

because conjugating both pairs of image coordinates by h permutes the constraints. The identity atom and the objective are unchanged. Thus a conjugation-invariant optimal pair exists. Such pairs assign a constant probability density to each conjugacy class, and their equality of one-point marginals is equivalent to the reduced moment equation $Mp = Mq$: for a central law, each entry $\Pr(gx = y)$ is constant on the orbital containing (x, y) , and its orbital average is $\mathbb{E}F_j/|\mathcal{O}_j|$. This proves the second line of (5).

Similarly, average any dual function ϕ over conjugations. The indicator $\mathbf{1}_{\{g=e\}}$ is conjugacy invariant, while oscillation is convex and invariant under conjugation; the average cannot increase oscillation. A conjugation-averaged linear combination of the image indicators has coefficients constant on their simultaneous G -orbits in $\Omega \times \Omega$. Therefore it lies in $\text{span}\{F_j\}$. This proves the fourth line of (5). Evaluating the same oscillation on conjugacy classes gives the third line, whose entries are precisely (3).

When the optimum $\gamma > 0$, an optimal reduced pair P, Q must be **mutually singular**. Otherwise $d = \|P - Q\|_{\text{TV}} < 1$, and its Jordan parts normalized by d would form an admissible pair with strictly larger objective $\gamma/d > \gamma$, contradiction. Thus the supports of an optimal central P, Q are disjoint and (6) is a probability measure for every $0 \leq \delta \leq (|G| \max_g Q(g))^{-1}$, with all the stated identities.

Finally, for any law ν with the uniform law's one-point marginals, $v = \nu - u_G \in \mathcal{V}$ and the defining norm bound gives (8) at the identity. Replacing ν by its left translate by σ^{-1} preserves marginal equality with u_G , total variation and atom excess, proving (8) for every σ . The same translation transports the attained equality. QED.

2 Context and scope

This theorem extracts the universal finite-action principle from Section 18 of the repository's sharp robust permanent note. The separate closed formulas for particular symmetric-group actions are outside this paper. The duality and rational-attainment inputs are classical finite-dimensional linear programming; see A. Schrijver, *Theory of Linear and Integer Programming*, Wiley, 1986. The formal proof supplies the required real-dual and rationalization bridges using proved finite-dimensional linear algebra and Hahn–Banach. No claim of resolving a famous open conjecture is made.

3 Formal verification and reproduction

The principal theorem has a complete Lean proof. The precise scope is: Only the universal finite-action theorem, rational primal/dual attainment, all-atom sharpness and degenerate zero-optimum case. Separate explicit symmetric-group formulas in the parent note are not certified by this paper. The public proof source snapshot is fixed at commit 02af7db99db789fecb0d5f5f8f37b1a4e40762a0 in <https://github.com/mxym/math>. The proof packages, with their own dependency locks, audit records and reproduction instructions, are:

- `formalizations/orbital-primal-dual`.

The main formal entry points are:

- `OrbitalMarginals.universal_orbital_theorem`.
- `OrbitalMarginals.orbital_span_exact_minimum`.
- `OrbitalMarginals.sharp_attainment_all_atoms`.

Lean is fixed to version 4.34.1, compiler commit 5045d0056413266e57c625dcd7c365b10e377c52; Mathlib is fixed to d13f23b723b8a846827a245b89c10fc7d3f11612. The audited mathematical roots depend only on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. The archived audits state their actual compilation and replay scopes; publication of this typeset version does not imply a new execution of every previous empty-kernel replay. The companion README maps each printed principal result to its formal theorem and explains how to repeat verification.

Research and preparation disclosures

This research, formalization and manuscript preparation used AI assistance. The author is responsible for the manuscript. The principal claims are supported by the readable proofs and the public Lean artifacts specified above. Formal verification is not external human peer review; no such review is claimed. No external funding was received. No worldwide priority or journal acceptance is asserted.